XIANMEI REN

Ph.D. Candidate in Statistics | Expected June 2027

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RESEARCH INTERESTS

Intelligent optimization

Evolutionary computation (Evolutionary algorithms and swarm intelligence)

Scheduling and optimization in manufacturing and transportation

Artificial intelligence (Reinforcement learning)

EDUCATION

Zhejiang University of Finance & Economics, China **2023–Present**
Ph.D. in Statistics

Dongbei University of Finance & Economics, China **2018–2020**
M.Sc. in Applied Statistics

Zaozhuang University, China **2013–2017**
B.Sc. in Applied Statistics

PUBLICATIONS

1. **Ren, X. M.**, Zhang, W. Y., Liu, X. Q., Li, Z. H., & He, Y. F. (2026). Digital twin-driven dynamic scheduling in flexible job shop considering worker factors towards Industry 5.0. *Expert Systems with Applications, 315, 131788*. *[Published]*
   https://doi.org/10.1016/j.eswa.2026.131788
2. **Ren, X. M.**, Zhang, S., & He, Y. F. Dynamic flexible job shop scheduling under worker constraints with learning-forgetting effects via Q-learning-enhanced algorithm assisted by digital twin. *International Journal of Production Economics*. *[Under revision]*
3. **Ren, X. M.**, Zhang, S., Zhang, W. Y., & Chen, Y. Inverse flexible job shop scheduling with transportation under coupled failures via improved decomposition-based evolutionary algorithm. *Computers & Industrial Engineering*. *[Under revision]*
4. Wu, Y., Wang, W. R., **Ren, X. M.** (corresponding author), Zheng, H., & Yuan, C. Y. X. A novel multi-visit truck-drone collaborative optimization model for the green vehicle routing problem with dynamic flight endurance. *Journal of Combinatorial Optimization*. *[Under revision]*

PATENT APPLICATIONS

1. Zhang, S., **Ren, X. M.**, Zhang, W. Y., Chen, Y., Nie, J. H., & Huang, Z. K. (2025). “A Digital Twin-Driven Dynamic Scheduling Method for Flexible Job Shops.” Chinese National Invention Patent Application No. 2025119440694. Filed 22 December 2025; under substantive examination.
2. Zhang, S., **Ren, X. M.**, Xu, H. F., & Zhang, W. Y. (2026). “A Robust Scheduling Method and Apparatus for Remanufacturing Job Shops Considering Carbon Emission Constraints.” Chinese National Invention Patent Application No. 202610348181X. Filed 20 March 2026; pending substantive examination.
3. Zhang, S., **Ren, X. M.**, Lu, G., Zhang, W. Y., Nie, J. H., & Chen, Y. (2026). “A Digital Twin-Based Dynamic Scheduling Method for Manufacturing Workshops Considering Learning and Forgetting Effects.” Chinese National Invention Patent Application No. 2026109973199. Filed 6 July 2026; pending substantive examination.

CONFERENCE PRESENTATION

2026 · 3rd International Conference on Materials Engineering & Intelligent Manufacturing

Oral presentation: *Digital Twin-Driven Dynamic Scheduling of Flexible Job Shops*

RESEARCH PROJECT

Graduate research project Zhejiang University of Finance & Economics

Role: Participant

Project title: Research on resource organization and dynamic optimization of post-disaster truck-drone collaborative emergency logistics empowered by the low-altitude economy

Duration: Jun. 2026 – May. 2027 (Ongoing) | Project No.: 26TYDC026

AWARDS

Second-Class Academic Scholarship **2023–2024;2024–2025**
Zhejiang University of Finance & Economics

Outstanding Graduate Student **2024–2025**
Zhejiang University of Finance & Economics

PROFESSIONAL & TEACHING EXPERIENCE

Shantui Construction Machinery Co., Ltd. *[Data Development Engineer]* **2020–2023**

- Developed data analysis and visualization solutions to support manufacturing operations and decision-making.
- Translated business requirements into analytical solutions in collaboration with cross-functional teams.

Zhejiang University of Finance & Economics *[Teaching Assistant]* **2023–Present**

- Supported course preparation, tutorials, student consultation, classroom discussions, and assessment.
- Assisted with seminar- and case-based teaching; guided students in practical coursework.